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$$2^{x-1} + 2^{1-x} = \frac{5}{2}$$

$$\Rightarrow \frac{2^x}{2} + \frac{2}{2^x} = \frac{5}{2}$$

Stel $t = 2^x$ met ($t > 0$):

$$\Rightarrow \frac{t}{2} + \frac{2}{t} = \frac{5}{2}$$

$$\Rightarrow t^2 + 4 = 5t$$

$$\Rightarrow t^2 - 5t + 4 = 0$$

$$\Rightarrow (t - 1)(t - 4) = 0$$

$$\Rightarrow t = 1 \text{ of } t = 4$$

$$t = 1 \Rightarrow 2^x = 1 \Rightarrow x = 0$$

$$t = 4 \Rightarrow 2^x = 4 \Rightarrow x = 2$$

$$x \in \{0, 2\}$$

References